MindBloom Learn™
Helping Every Mind Bloom!

STEMQuest™ ACTIVITIES

Science • Technology • Engineering • Math

AGES 4–8

Let's celebrate Independence Day with fun STEM activities!

JULY 4th STEM CHALLENGE SHEET

A Fun Way to Explore, Create & Learn!

1 Rocket Countdown

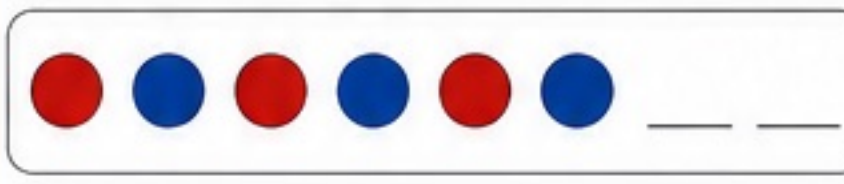
Help Nova count backwards.

10, 9, 8, ___, ___, ___,

___, ___, ___, ___

2 Build a Firework Pattern

What comes next?



3 Count the Fireworks

How many fireworks do you see?

Answer: ________

How many stars?

Answer: ________

4 Which Launches Higher?

Circle the object that can fly highest.

Kite Rocket Bicycle

5 Red, White & Blue Sort

Draw a line to the correct color group.

RED

WHITE

BLUE

YELLOW

6 STEM Observation Challenge

Look at the picture. Count.

Stars ________

Fireworks ______

Flags ________

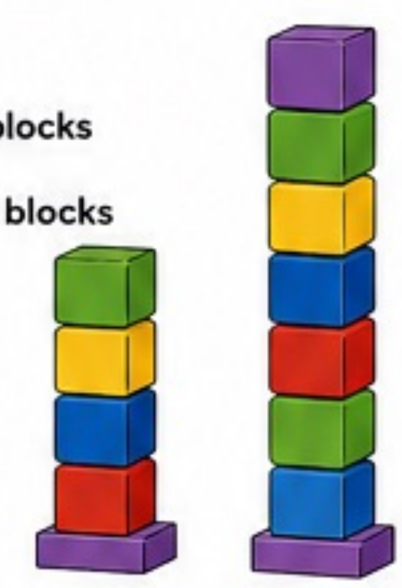
7 Build a Tower Challenge

Using blocks:

- ☐ Build a tower with 5 blocks
- ☐ Build a tower with 10 blocks

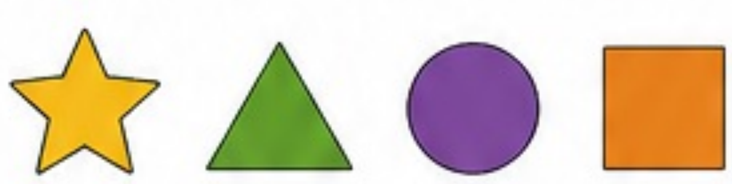
Which tower is taller?

____



8 Firework Shapes

Circle the shape.



Star Triangle Circle Square

Which shape looks most like a firework?

Answer: ____________________

9 Sink or Float Prediction

Predict before testing.

Item	Sink	Float
Cork	☐	☐
Coin	☐	☐
Pencil	☐	☐
Leaf	☐	☐

10 Nova's Mini Engineer Challenge

Can you build a paper rocket?
Draw your rocket below.

BONUS CHALLENGE

How many stars are on the American flag?

- ○ 25
- ○ 50
- ○ 100

Answer: ________

SKILLS YOU'LL BUILD

- Counting
- Prediction Skills
- Observation
- Early STEM Concepts
- Pattern Recognition
- Problem Solving
- Engineering Thinking

NOVA'S STEM TIP

Scientists ask questions.
Engineers build solutions.
Today you can be both!

Name: ________________________ Date: __________________

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JULY 4th ENGINEERING & SCIENCE CHALLENGE SHEET

AGES 9–12

AGES 9–12

Think. Solve. Create. Celebrate Freedom!

1 BUILD A FIREWORK LAUNCHER

You have 10 craft sticks, 3 rubber bands, and a plastic spoon. Design a launcher that can launch a pom-pom as far as possible!

Draw your design below and label the parts.

How do you think your launcher works?

____

____

2 FIREWORKS MATH

Solve the problems below.

a) 489 + 276 = ________

b) 1,250 – 437 = ________

c) (213 × 5) + 125 = ________

d) 2,016 ÷ 8 = ________

e) A show uses 7 rows of 9 fireworks. How many fireworks are in the show? ________

f) If each firework lasts 8 seconds, how long will the whole show last?

____

3 ENGINEERING DESIGN CHALLENGE

Design a safe and eco-friendly firework that creates a beautiful effect.

Fill in the design planner below.

Shape of Firework: ________________

Colors Used: ________________

Effect: ________________

Materials (safe & eco-friendly): ________________

____

How is it eco-friendly? ________________

____

4 DECODE THE SECRET MESSAGE

Use the key to decode the message.

★	○	🇺🇸	۞	♥	!
A	E	I	O	U	!

★ ○ 🇺🇸 ★ ○ ♥ !
___ ___ ___ ___ ___ ___ ___

★ 🇺🇸 ○ ○ ★ !
___ ___ ___ ___ ___ ___

○ ۞ ★ ○ 🇺🇸 ★ !
___ ___ ___ ___ ___ ___ ___

★ ○ ♥ ★ 🇺🇸 !
___ ___ ___ ___ ___ ___

5 PATTERN LOGIC

What comes next?

1) 🔴 🟦 ⚪ 🔴 🟦 ⚪ ____ ____

2) ★ ★ ☆ ★ ★ ☆ ★ ____ ____

3) 2, 4, 8, 16, 32, ______, ______

4) 🇺🇸 🔴 🇺🇸 🔵 🇺🇸 🔴 ______, ______

5) 1, 3, 6, 10, 15, ______, ______

6 DATA IN ACTION

Look at the chart and answer the questions.

Fireworks Sold at the Festival

Type	Friday	Saturday	Sunday	Total
Rockets	36	54	42	____
Fountains	28	32	25	____
Sparklers	45	60	50	____
Total	____	____	____	____

a) Which firework type was sold the most on Saturday? ________

b) How many rockets were sold in total? ________

c) How many more sparklers were sold than fountains on Sunday? ________

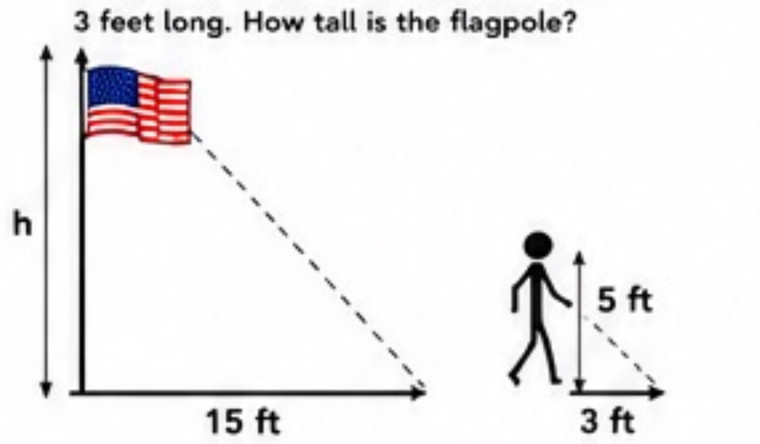
7 STEM TRIVIA

Circle the best answer.

1. What gas helps fireworks create colors?
   A) Oxygen B) Helium C) Nitrogen D) Carbon Dioxide
2. Which law helps rockets move upward?
   A) Law of Gravity B) Newton's First Law
   C) Newton's Third Law D) Ohm's Law
3. What unit measures force?
   A) Watt B) Joule C) Newton D) Liter
4. What is the main purpose of a fuse in fireworks?
   A) To make sound
   B) To control when it explodes
   C) To change the color
   D) To make it bigger

8 MEASUREMENT CHALLENGE

A flagpole casts a shadow 15 feet long at the same time a person who is 5 feet tall casts a shadow 3 feet long. How tall is the flagpole?



Flagpole height = ____________ feet

9 SIMPLE CIRCUIT CHALLENGE

Draw a simple circuit to light up a LED (light-emitting diode). Use a battery, wires, and an LED.

What happens if the wires are not connected correctly?

____

10 INVENT INDEPENDENCE DAY 2050

Imagine it's the year 2050! How will we celebrate Independence Day using technology?
Draw or describe your ideas below.

____

____

"The future belongs to those who believe in the beauty of their dreams."
– Eleanor Roosevelt

NOVA'S STEM TIP

Engineers solve problems, try new ideas, and never give up. You can be an engineer too!

SKILLS YOU'LL BUILD

Problem Solving | Math Skills | Engineering Design | Data Analysis | Creativity & Innovation | Critical Thinking

NOVA'S MISSION

Inspire young minds to explore, innovate, and build a brighter future!

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JULY 4th ENGINEERING & SCIENCE CHALLENGE SHEET

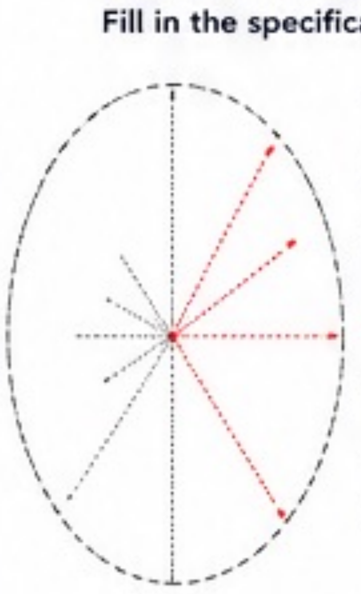
AGES 13–19 TEENS

Think. Solve. Innovate. Celebrate Freedom!

AGES 13–19 TEENS

1 FIREWORKS ENGINEERING

Design your own aerial firework shell.
Fill in the specifications for your design.



Shell Diameter: ________ cm

Burst Shape: ______________

Primary Color(s): ______________

Secondary Color(s): ______________

Estimated Launch Height: __________ feet

Special Effect: ______________

____

2 FIREWORKS MATH CHALLENGE

Solve the problems below.

a) 1,256 + 347 = __________

b) 2,480 – 965 = __________

c) (325 × 4) + 150 = __________

d) 3,600 ÷ 8 = __________

e) A firework kit has 192 rockets. You use $^3/_8$ of them. How many are left? __________

3 ENGINEERING DESIGN CHALLENGE

You have $50 to design a safe and exciting firework show.

List the materials you will use and the purpose of each.

Material	Cost	Purpose

Total Cost: $ ____________

4 DECODE THE SECRET MESSAGE

Use the key to decode the message.

★	○	🇺🇸	✺	♥	!
A	E	I	O	U	!

★ ○ 🇺🇸 ★ ○ ✺ !

★ ○ ♥ ★ 🇺🇸

★ ○ ♥ ✺ ○ 🇺🇸 ★ ★ !

5 PATTERN LOGIC

What comes next in each sequence?

1) 🔴 🔵 🔵 🔴 🔵 🔵 🔴 _____ _____

2) ★ ★ ☆ ★ ☆ ☆ ★ _____ _____

3) 🚀 🔵 🚀 🔵 🚀 🔵 _____ _____

4) 2, 6, 12, 20, 30, _____, _____

5) 1, 4, 9, 16, 25, _____, _____

6 DATA ANALYSIS

The table shows the number of fireworks launched in a show.

Type	Red	Blue	Gold	Total
Shells	24	18	12	_____
Rockets	15	10	25	_____
Sparklers	20	15	10	_____
Total	_____	_____	_____	_____

a) Which type is used the most? ____________

b) How many gold fireworks in total? ____________

c) If 20 more blue shells are added, what is the new total for blue? ____________

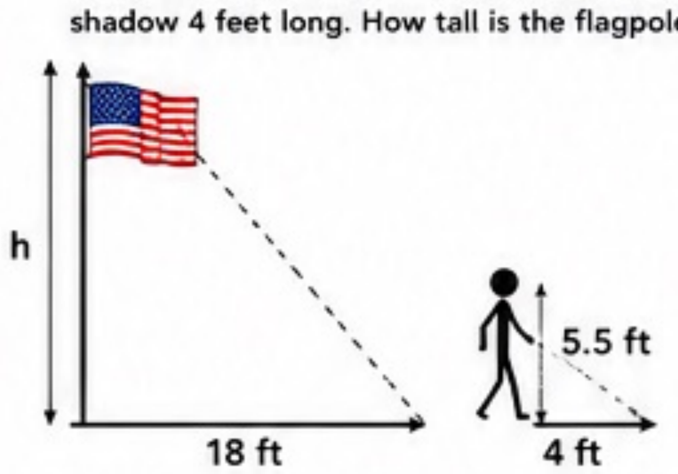
7 STEM TRIVIA

Circle the best answer.

1. What gas gives fireworks their bright colors?
   A) Oxygen B) Helium C) Copper D) Carbon Dioxide
2. Which law is used when a rocket is launched?
   A) Newton's First Law B) Newton's Second Law
   C) Newton's Third Law D) Law of Gravity
3. What unit measures power of an engine?
   A) Joule B) Newton C) Watt D) Liter
4. What is the main purpose of a fuse in fireworks?
   A) To make sound
   B) To control the timing of the explosion
   C) To change the color
   D) To increase the size

8 MEASUREMENT CHALLENGE

A flagpole casts a shadow 18 feet long at the same time a person who is 5.5 feet tall casts a shadow 4 feet long. How tall is the flagpole?



Flagpole height = ____________ feet

9 PARADE FLOAT BLUEPRINT

Design a parade float that represents freedom and innovation.
Sketch your design below. Label the key parts and materials you would use.

Materials: ______________________

____

10 INVENT INDEPENDENCE DAY 2050

Imagine the celebrations of the future!

Describe or draw technologies, innovations, or ideas that could make Independence Day even better, safer, and more meaningful in the year 2050.
What will people see, do, and experience?

____

____

____

> "The best way to predict the future is to create it."
> – Peter Drucker

NOVA'S STEM TIP

Great engineers ask questions, test ideas, and never stop improving. Keep challenging your mind every day!

SKILLS YOU'LL BUILD

Critical Thinking · Problem Solving · Math Skills · Engineering Design · Data Analysis · Creativity & Innovation · Communication & Planning

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Inspire young minds to explore, innovate, and build a brighter future!